

Exercise 3: Loops, Conditionals, and the Empty State

Context: You have a single card working. Now you need a catalog of courses, and you must explicitly handle the case where there are no courses available using Angular's modern built-in control flow blocks.

[!NOTE]

Step 1: Create the Catalog Dataset

Open `src/app/features/student-dashboard/student-dashboard.component.ts`.
Replace your singular `sampleCourse` property with a reactive array signal:

```
availableCourses = signal<Course[]>([
  {
    id: 1,
    title: "Advanced Java Services",
    code: "CSE-101",
    maxCapacity: 30,
    enrollmentCount: 10,
  },
  {
    id: 2,
    title: "Angular UI Lab",
    code: "CSE-210",
    maxCapacity: 25,
    enrollmentCount: 25,
  },
  {
    id: 3,
    title: "Database Design",
    code: "CSE-305",
    maxCapacity: 20,
    enrollmentCount: 18,
  },
  {
    id: 4,
    title: "API Security Workshop",
    code: "CSE-420",
    maxCapacity: 40,
    enrollmentCount: 15,
  },
]);
```

[!NOTE]

Step 2: Render the Control Flow Template Loop

Required Cleanup: Exercise 2 left a single `<tms-course-card>` bound to `sampleCourse`.
Remove that extra `<h2>` and the single `<tms-course-card [course]="sampleCourse"`

.../> from your HTML file to prevent redundant rendering. You can safely delete the unused `sampleCourse` property from your TypeScript class once no code references it.

Open `src/app/features/student-dashboard/student-dashboard.component.html`.

Replace the old catalog markup with Angular's modern structural control blocks:

```
<h2>Course Catalog</h2>

@if (availableCourses().length === 0) {
  <div class="empty-state">
    <p>No courses are available this term. Check back during
registration.</p>
  </div>
} @else {
  <div class="grid">
    @for (course of availableCourses(); track course.id) {
      <tms-course-card [course]="course"
(enrollClicked)="handleEnroll(\$event)" />
    } @empty {
      <p>No results match your search.</p>
    }
  </div>
}

@if (selectedCourse(); as picked) {
  <p class="selection-hint" role="status">
    Last enrollment request: <strong>{{ picked.title }}</strong>
    ({{ picked.code }})
  </p>
}
```

Core Declarative Syntax Rules Explained:

- `track course.id`: This statement is **mandatory** inside the `@for` block. It establishes a unique identity anchor for every item, allowing Angular to track DOM nodes efficiently during list modifications. Omitting the `track` parameter causes a compilation error.
- `@empty`: A built-in sub-block that renders automatically if the collection array evaluated by the `@for` loop contains zero elements.
- `selectedCourse(); as picked`: Evaluates the signal and assigns its current non-null value to a local template variable (`picked`). This visual feedback explicitly proves the parent component successfully caught the emitted child event.